DERIVED HOM COMMUTES WITH THE LOCALIZATION FUNCTOR FOR PSEUDO-COHERENT OR PERFECT FIRST ARGUMENT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-perfect.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: both cases are proved by explicit complex representations and reduce to the same finite-free pointwise isomorphism; the argument is complete

Editorial note. The label lemma-descend-RHom names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- the localization functor epsilon^* from O_X-modules to O_etale-modules on the small etale site and its flatness
- RSheafHom in the derived category of modules on a ringed site
- pseudo-coherent and perfect complexes represented by finite free resp. strictly perfect complexes
- the Rhom representation lemmas for complexes of direct summands of finite free modules and for strictly perfect complexes

Source reference. Original lemma 08I5: sources/stacks-chow/tags.txt, source lines 10803–10803. Prerequisite pointer invoked at proof line(s) 3456; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 08DM: sources/stacks-chow/tags.txt, source lines 10642–10642. Prerequisite pointer invoked at proof line(s) 3495; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original equation 08H7: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 320–320. Prerequisite pointer invoked at proof line(s) 3437; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 08H8: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 330–330. Prerequisite pointer invoked at proof line(s) 3441; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:spaces-perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 06Z0: sources/stacks-chow/tags.txt, source lines 8904–8904. Prerequisite pointer invoked at proof line(s) 3486; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 08EE: sources/stacks-chow/tags.txt, source lines 10670–10670. Prerequisite pointer invoked at proof line(s) 3452; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 08EB: sources/stacks-chow/tags.txt, source lines 10667–10667. Prerequisite pointer invoked at proof line(s) 3487; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 08JI: sources/stacks-chow/tags.txt, source lines 10851–10851. Prerequisite pointer invoked at proof line(s) 3463; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original remark 08JF: sources/stacks-chow/tags.txt, source lines 10848–10848. Prerequisite pointer invoked at proof line(s) 3445; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let X be a scheme. Let E, F be objects of $D(\mathcal{O}_X)$. Assume either*

(1) *E is pseudo-coherent and F lies in $D^+(\mathcal{O}_X)$, or*
(2) *E is perfect and F arbitrary,*

then there is a canonical isomorphism

$$\epsilon^* R\mathcal{H}om(E, F) \longrightarrow R\mathcal{H}om(\epsilon^* E, \epsilon^* F)$$

Here ϵ is as in (08H7).

Proof. Recall that ϵ is flat (Lemma 08H8) and hence $\epsilon^* = L\epsilon^*$. There is a canonical map from left to right by Cohomology on Sites, Remark 08JF. To see this is an isomorphism we can work locally, i.e., we may assume X is an affine scheme.

In case (1) we can represent E by a bounded above complex $\mathcal{E}^\bullet$ of finite free $\mathcal{O}_X$-modules, see Derived Categories of Schemes, Lemma 08EE. We may also represent F by a bounded below complex $\mathcal{F}^\bullet$ of $\mathcal{O}_X$-modules. Applying Cohomology, Lemma 08I5 we see that $R\mathcal{H}om(E, F)$ is represented by the complex with terms

$$\bigoplus\nolimits_{n=-p+q} \mathcal{H}om_{\mathcal{O}_X}(\mathcal{E}^p, \mathcal{F}^q)$$

Applying Cohomology on Sites, Lemma 08JI we see that $R\mathcal{H}om(\epsilon^* E, \epsilon^* F)$ is represented by the complex with terms

$$\bigoplus\nolimits_{n=-p+q} \mathcal{H}om_{\mathcal{O}_{étale}}(\epsilon^* \mathcal{E}^p, \epsilon^* \mathcal{F}^q)$$

Thus the statement of the lemma boils down to the true fact that the canonical map

$$\epsilon^* \mathcal{H}om_{\mathcal{O}_X}(\mathcal{E}, \mathcal{F}) \longrightarrow \mathcal{H}om_{\mathcal{O}_{étale}}(\epsilon^* \mathcal{E}, \epsilon^* \mathcal{F})$$

is an isomorphism for any $\mathcal{O}_X$-module $\mathcal{F}$ and finite free $\mathcal{O}_X$-module $\mathcal{E}$.

In case (2) we can represent E by a strictly perfect complex $\mathcal{E}^\bullet$ of $\mathcal{O}_X$-modules, use Derived Categories of Schemes, Lemmas 06Z0 and 08EB and the fact that a perfect complex of modules is represented by a finite complex of finite projective modules. Thus we can do the exact same proof as above, replacing the reference to Cohomology, Lemma 08I5 by a reference to Cohomology, Lemma 08DM. □